A Simplified FSO Channel Model with Weak Turbulence and Pointing Errors

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Student assignment — solution and follow-up proposal
Paper: C. Álvarez Roa et al., ICTON 2024

Channel modelling • Numerical stability • Monte Carlo • Outage • Interleaving

The Channel Model: Three Multiplicative Losses



$$y = \eta\, h\, x + n, \qquad h = h_l\, h_p\, h_a, \qquad x \in \{0, 2P\}, \qquad n \sim \mathcal{N}(0, \sigma_n^2)$$

Path loss h_l

Deterministic

- Beer–Lambert law:
  $h_l = e^{-\sigma(\lambda) L}$
- With Table I:
  $h_l = 10^{-\frac{0.2208 \times 3}{10}} = 0.859$
- $\sigma(\lambda)$ changes over minutes–hours
  $\Rightarrow$ constant in the analysis window

Spread + pointing h_p

Random

- $w_z \approx \theta_{1/2} L = 1.98\,\mathrm{m} \gg a = 5\,\mathrm{cm}$
  $A_0 = \mathrm{erf}^2(v) = 1.27 \times 10^{-3}$
- Jitter $x, y \sim \mathcal{N}(0, \sigma_s^2) \Rightarrow R$ Rayleigh:
  $h_p = A_0\, e^{-2R^2/w_{z,eq}^2}$
- $f_{H_p}(h_p) = \dfrac{\gamma^2}{A_0^{\gamma^2}}\, h_p^{\gamma^2 - 1}, \quad \gamma = \dfrac{w_{z,eq}}{2\sigma_s}$

Turbulence h_a

Random

- $h_a = e^{2\chi}, \;\; \chi \sim \mathcal{N}(-\sigma_X^2, \sigma_X^2)$, so that $\mathbb{E}[h_a] = 1$
- $f_{H_a}(h_a) = \dfrac{1}{2h_a\sqrt{2\pi\sigma_X^2}} \exp\!\left(-\dfrac{(\ln h_a + 2\sigma_X^2)^2}{8\sigma_X^2}\right)$
- $\sigma_X^2 \approx \sigma_R^2/4,$
  $\sigma_R^2 = 1.23\, C_n^2\, k^{7/6} L^{11/6}$

Independent effects on optical power $\Rightarrow$ they multiply. Fading (h_p, h_a) changes over milliseconds $\gg$ nanosecond symbols.

One Change of Variable Explains the Model

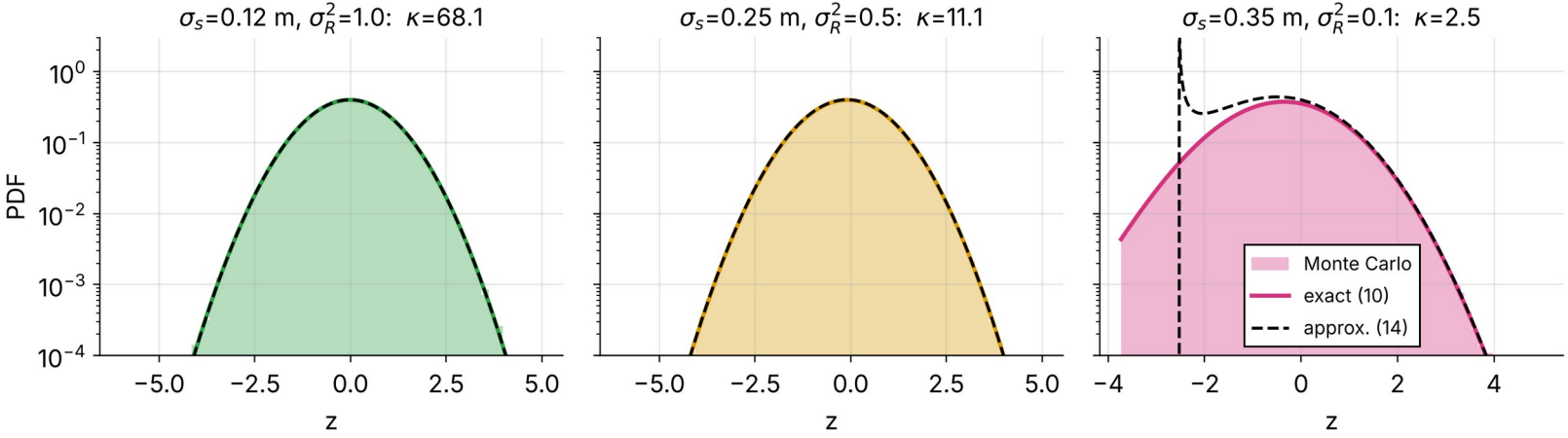
$$z = \frac{\ln \frac{h}{A_0 h_l} + 2\sigma_X^2}{2\sigma_X} = W - E', \qquad W \sim \mathcal{N}(0,1), \quad E' \sim \mathrm{Exp}(\kappa), \quad \kappa = \gamma^2 \sigma_R$$

Exact composite PDF, eq. (10)

$$f_Z(z) = \kappa\, e^{\kappa z + \kappa^2/2}\, Q(z+\kappa)$$

Approximation, eq. (14)

$$f_Z(z) \approx \frac{\phi(z)}{1 + z/\kappa}, \qquad z > -\kappa \;\; (h > \hat{h})$$



Why Eq. (10) Breaks for Small σ_s, and the Fix

$$\underbrace{e^{2\sigma_X^2\gamma^2(1+\gamma^2)}}_{e^{2353}} \cdot \underbrace{(A_0 h_l)^{-\gamma^2}}_{e^{464}} \cdot \underbrace{h^{\gamma^2-1}}_{e^{-458}} \cdot \underbrace{\operatorname{erfc}(\cdot)}_{e^{-2358}} \;\to\; \infty \cdot 0 = \mathrm{NaN}$$

Cause

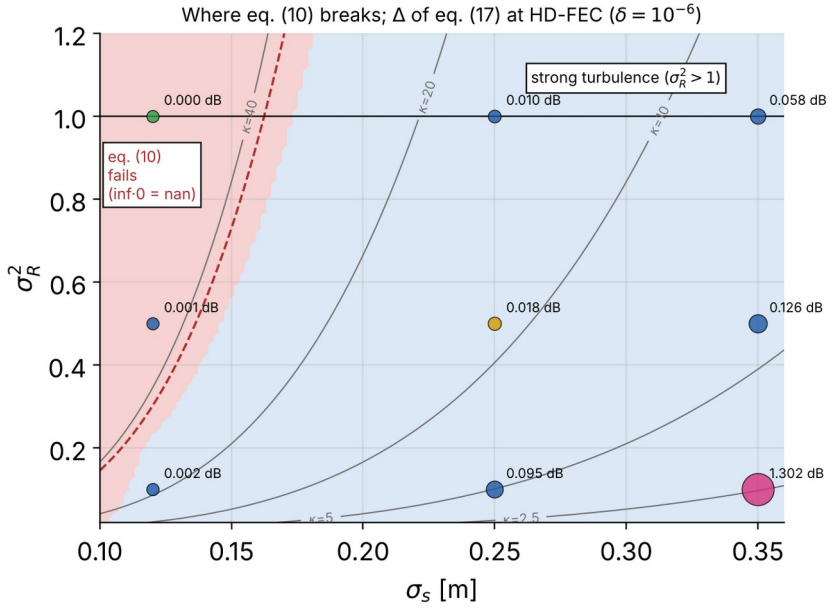
- Green point: $\gamma^2 = 68.1$, $\kappa = 68.1$; double precision holds only $e^{-745} \ldots e^{709.8}$
- Largest exponent $\approx 2\sigma_X^2\gamma^4 = \kappa^2/2$: small $\sigma_s \Rightarrow$ large γ
- Fails when $2\sigma_X^2\gamma^2(1+\gamma^2) > 709.8$

Fix

Add the exponents first, then use a log-Q routine:

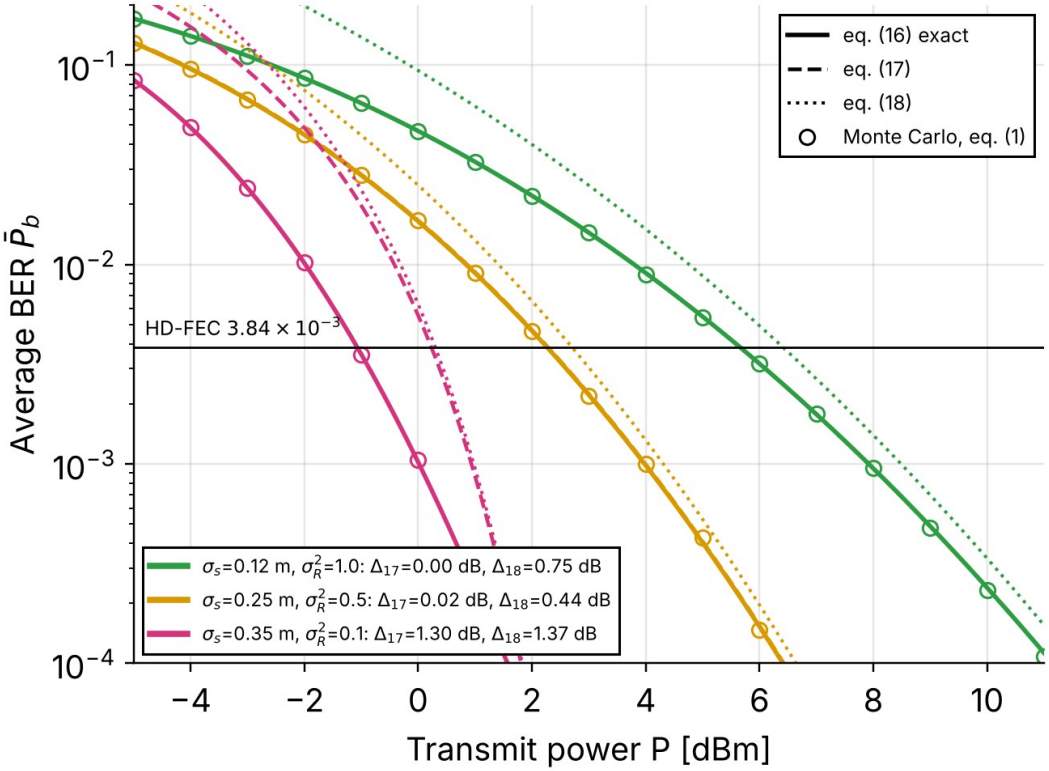
$$\ln f_Z = \ln\kappa + \kappa z + \frac{\kappa^2}{2} + \ln Q(z+\kappa)$$

Exact and stable for every (σ_s, σ_R^2).



Red: literal (10) fails. Dashed: overflow limit. Grey: contours of κ

Fig. 2 Reproduced: Monte Carlo, Exact (16), (17), (18)

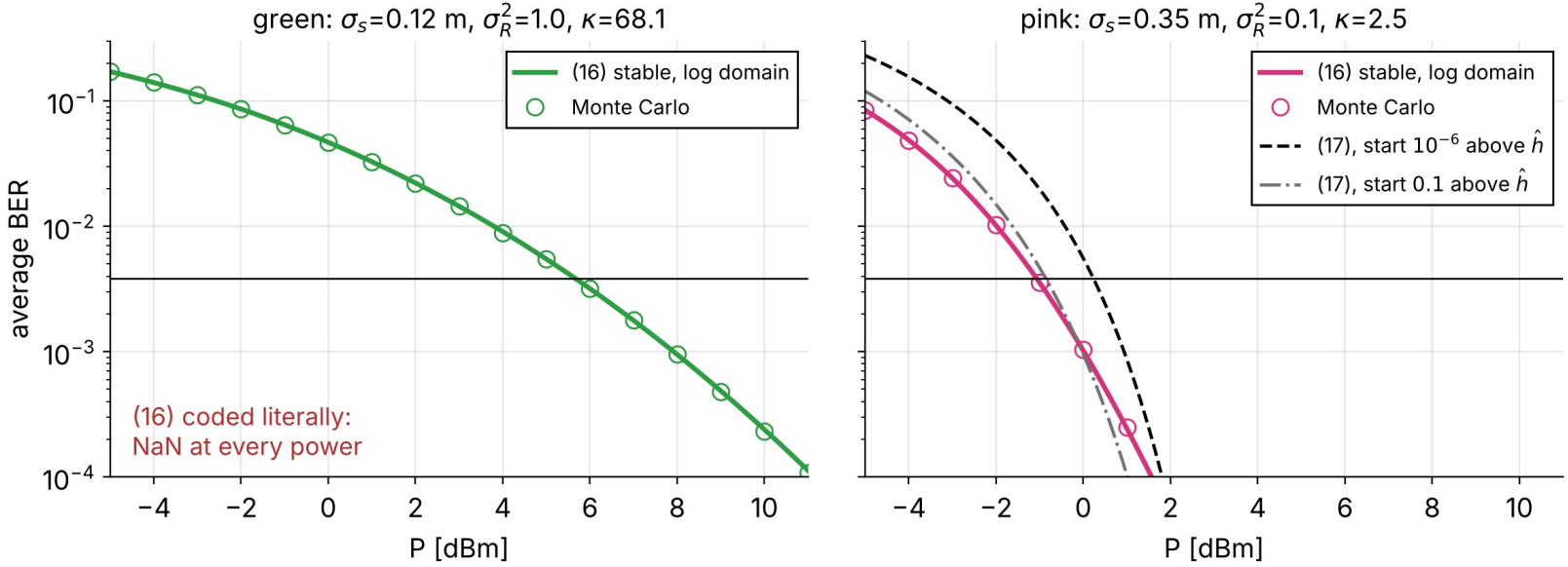


Point	Δ_{17} ours	Δ_{17} paper	Δ_{18} ours
green	0.0004	≈ 0	0.75
yellow	0.018	0.02	0.44
pink	1.30*	0.18	1.37

*Δ in dB at the HD-FEC limit 3.84·10⁻³. *Depends on the lower integration limit (next slide).*

- Δ_{17} within 0.015 dB of the paper at 8 of 9 points
- Monte Carlo (4×10^6 bits/point) within 2.7σ of exact (16) at 40 of 41 powers
- (18) is 0.4–0.8 dB pessimistic, as in the paper

Two Cases the Paper Could Not Resolve



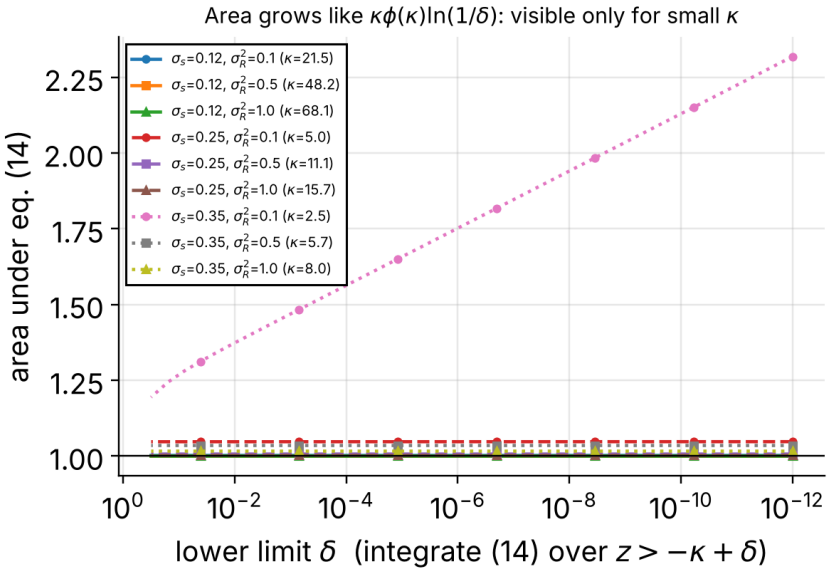
Green: σ_s = 0.12 m, σ_R^2 = 1

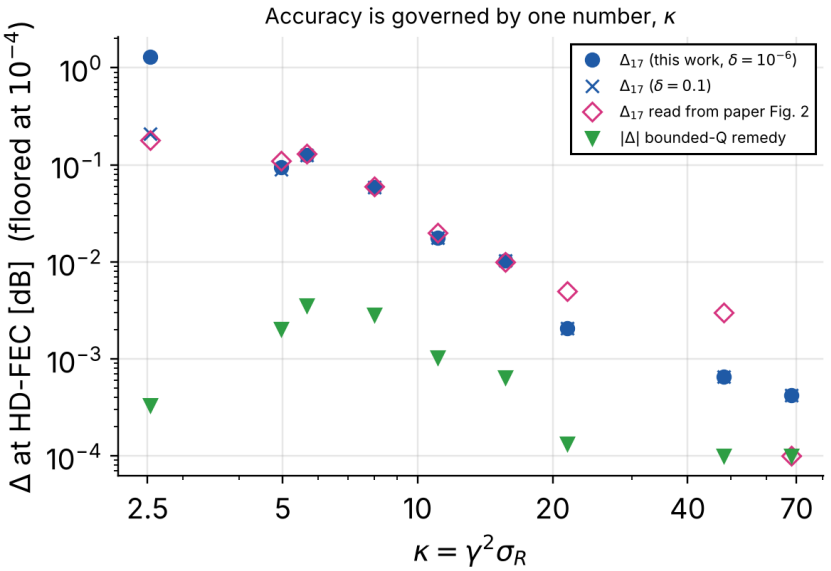
(16) coded literally: $e^{2353} \cdot 0 = \mathrm{NaN}$ at every P.
In log form it is finite and lies on the Monte Carlo points.

Pink: σ_s = 0.35 m, σ_R^2 = 0.1

(14) has a pole: $\int_{-\kappa+\delta} \frac{\kappa\phi(z)}{z+\kappa}\, dz \sim \kappa\phi(\kappa) \ln \frac{1}{\delta}$.
$\Delta_{17} = 0.21$ dB ($\delta = 0.1$) $\ldots$ 1.30 dB ($\delta = 10^{-6}$).

Task 3: Area Under (14) and Accuracy Are Set by κ



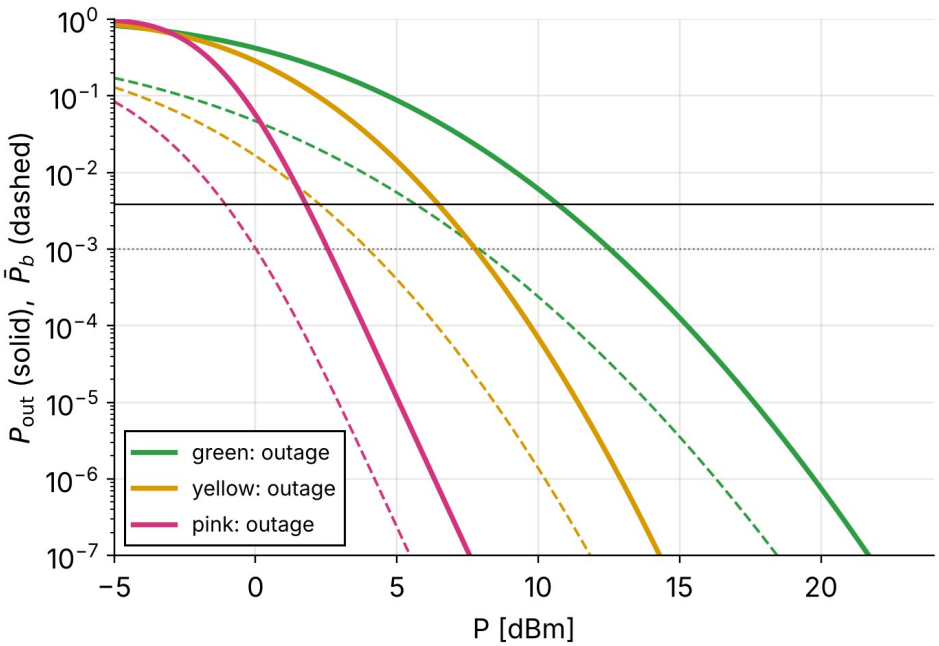


Area under eq. (14)

$$\text{area}(14) \approx 1 + \frac{1}{\kappa^2} + \frac{3}{\kappa^4} + \kappa\phi(\kappa)\ln\frac{1}{\delta}$$

- erfc argument $\approx \kappa/\sqrt{2} \Rightarrow$ error of (13) $\approx 1/\kappa^2$
- More turbulence or less jitter $\Rightarrow$ larger $\kappa = \gamma^2\sigma_R$
- Bounded Q (Börjesson–Sundberg): $|\Delta| < 0.004$ dB at all 9 points

Stretch (b): A Coded Link Sees Outage, Not Average BER



Time scales

$R_s T_c = 10^{10}\,\text{Bd} \times (1\text{–}5)\,\text{ms} = 10^7\text{–}5 \times 10^7$ symbols per fade $\gg$ codeword $\Rightarrow$ each codeword sees one h.

Outage probability, closed form

$$h_{\text{th}} = \frac{\sqrt{2}\,\sigma_n\,\text{erfc}^{-1}(2\beta)}{\eta P}, \qquad \beta = 3.84 \times 10^{-3}$$

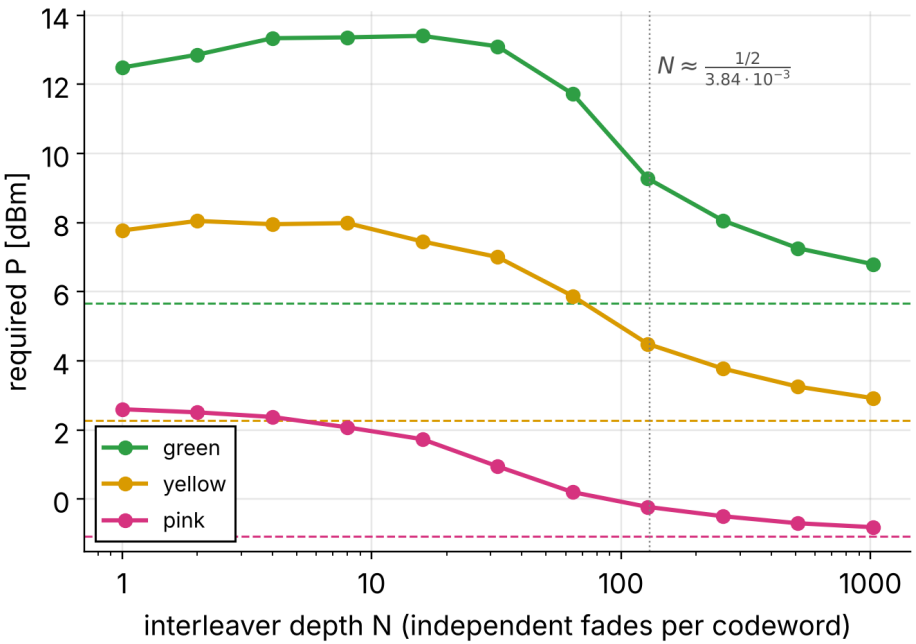
$$P_{\text{out}} = \Pr\{h < h_{\text{th}}\} = \Phi(z_{\text{th}}) + e^{\kappa z_{\text{th}} + \kappa^2/2}\, Q(z_{\text{th}} + \kappa)$$

Extra power for $P_{out} = 10^{-3}$ vs. average-BER design

6.9 dB	5.5 dB	3.6 dB
green	yellow	pink

A decoder succeeds or fails per codeword: the average BER mixes good blocks with rare BER ≈ ½ fades that no FEC can repair.

What Next: How Deep Must the Interleaver Be?



Proposal

Spread each codeword over N independent fades (latency $\sim NT_c$, memory NR_sT_c) and find the power for $P_{\text{fail}}(N) = \Pr\left\{\frac{1}{N}\sum_{i=1}^{N} P_b(h_i) > \beta\right\} = 10^{-3}$.

Preliminary result (Monte Carlo)

- $\frac{1}{2N} > \beta \iff N < 130$: one fade with $P_b \approx \frac{1}{2}$ fails the codeword; small N does not help
- $N = 1024$ (≈ 1 s at $T_c = 1$ ms): still 1.1 dB from $N \to \infty$ (green)

Proposed analysis (to be derived)

$$Y = P_b(H),\ M_Y(s) = \mathbb{E}\left[e^{sY}\right] = \int e^{sP_b(h(z))} f_Z(z)\, dz$$

$$P_{\text{fail}}(N) \le \inf_{s>0} \left[e^{-s\beta} M_Y(s)\right]^N \quad \text{(Chernoff)}$$

Why it matters: FSO and satellite links have millisecond fades and a latency budget, so they operate at finite N, where the average-BER design is optimistic.

Summary

Model

$z = W - E',\ \kappa = \gamma^2 \sigma_R$

(10): $\kappa e^{\kappa z + \kappa^2/2} Q(z + \kappa)$

(14): $\phi(z)/(1 + z/\kappa)$

Numerics

(10) fails as $\infty \cdot 0$ for small σ_s.

In log form it is exact everywhere and matches Monte Carlo.

Fig. 2

Δ_{17} within 0.015 dB at 8 of 9 points.

Pink ($\kappa = 2.5$): (17) diverges at $\hat{h}$.

Accuracy

Error of (13) $\approx 1/\kappa^2$: improves with κ.

Bounded Q: $|\Delta| < 0.004$ dB everywhere.

Coded links and next step

Outage is closed form; $+3.6$–6.9 dB over the average-BER design ($P_{\text{out}} = 10^{-3}$).

Next: finite interleaver depth N.

> Every figure and number in this talk is reproduced by `python run_all.py`.

Thank you

Questions and discussion are welcome

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Code: github.com/sweety-w-sys/Student-Assignment-free-space-optics-with-turbulence-and-pointing-errors